RingMoGPT: A Unified Remote Sensing Foundation Model for Vision, Language, and Grounded Tasks

Peijin Wang, *Graduate Student Member, IEEE*, Huiyang Hu, *Graduate Student Member, IEEE*, Boyuan Tong, *Graduate Student Member, IEEE*, Ziqi Zhang, Fanglong Yao, *Member, IEEE*, Yingchao Feng, *Member, IEEE*, Zining Zhu, *Graduate Student Member, IEEE*, Hao Chang, *Graduate Student Member, IEEE*, Wenhui Diao, *Member, IEEE*, Qixiang Ye, *Senior Member, IEEE*, and Xian Sun, *Senior Member, IEEE*

***Abstract*— Recently, multimodal large language models (MLLMs) have shown excellent reasoning capabilities in various fields. Most of the existing remote sensing (RS) MLLMs solve image-level text generation problems (e.g., image captioning), but ignore the core issues of object-level recognition, location, and multitemporal changes in the field of RS. In this article, we propose RingMoGPT, a multimodal foundation model that unifies vision, language, and localization. Based on the idea of domain adaption, RingMoGPT can complete training by fine-tuning only a few parameters. To make the model capable of object detection and change captioning, we further propose a location- and instruction-aware querying transformer (Q-Former) and a change detection module, respectively. To improve the performance of RingMoGPT, we carefully design the pretraining dataset and the instruction-tuning dataset. The pretraining dataset contains over a half million high-quality image and text pairs, which are generated through a low-cost and efficient data generation paradigm. The instruction-tuning dataset contains more than 1.6 million question-answer pairs, including six downstream tasks: scene classification, object detection, visual question answering (VQA), image captioning, grounded image captioning, and change captioning. Our experiments show that RingMoGPT performs well on six tasks, especially its ability to analyze multitemporal data changes and identify dense objects. We also verified the model under a zero-shot setting, and the results show that the proposed RingMoGPT also has good generalization ability in the face of new data.**

***Index Terms*— Foundation model, instruction-tuning, multimodal large language model (MLLM), pretraining, remote sensing (RS).**

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Peijin Wang, Boyuan Tong, Zining Zhu, Hao Chang, and Xian Sun are with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: wangpeijin17@mails.ucas.ac.cn; Tongboyuan22@mails.ucas.ac.cn; zhuzining20@mails.ucas.ac.cn; changhao21@mails.ucas.ac.cn; sunxian@aircas.ac.cn).

Huiyang Hu is with the Key Laboratory of Network Information System Technology (NIST), Key Laboratory of Target Cognition and Application Technology (TCAT), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: huhuiyang22@mails.ucas.ac.cn).

Ziqi Zhang is with the State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China (e-mail: zhangziqi2017@ia.ac.cn).

Fanglong Yao, Yingchao Feng, and Wenhui Diao are with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: yaofanglong17@mails.ucas.ac.cn; fengyc@aircas.ac.cn; diaowh@aircas.ac.cn).

Qixiang Ye is with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: qxye@ucas.ac.cn).

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I. Introduction

In THE field of Earth observation, there has been a growing interest in the development of foundation models [1], [2], [3], [4], [5], [6]. However, most of these models primarily focus on visual recognition tasks, emphasizing the learning of image-level visual features while overlooking text-based semantic understanding and lacking robust reasoning capabilities. Furthermore, these models often rely on labeled datasets and require retraining for each downstream task, which restricts their ability to generalize effectively to unseen data.

A large number of related studies have demonstrated that the fusion of vision and language greatly enhances the interpretative capabilities of remote sensing (RS) models [7], [8], [9], [10], [11]. However, current models are often tailored to specific downstream tasks, with their functionality limited to primary image–text alignment and text generation. This limitation makes them less effective for more complex tasks, such as object recognition and change detection in RS. The advent of large-scale language models like GPT-4 [12] has driven the development of multimodal large language models (MLLMs), which integrate visual and language information. MLLMs have shown great potential across various downstream tasks, including image captioning, image retrieval, and visual question answering (VQA). These models excel in text comprehension and generation, opening up new opportunities for advancing research in RS.

Existing MLLMs can generally be divided into two categories. The first category includes foundation models

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pretrained from scratch with all parameters. For instance, the ImageBind model [13] is trained on billions of multimodal data pairs. While these models deliver strong performance, they demand substantial data and computational resources. The second category involves freezing the visual and large language model (LLM) parameters, training only adapter parameters. Examples include models like InstructBLIP [14] and LLaVA [15]. This approach, compared to fully parameterized models, requires fewer data samples while maintaining competitive accuracy. We argue that the latter method is more practical, as it not only reduces the number of training parameters but also provides flexibility for a wide range of tasks by adjusting the output format of the LLM.

In this article, our research focuses on developing a unified RS foundation model that integrates vision, language, and grounded tasks. It not only requires the model to have cognitive capabilities for general visual tasks, but also emphasizes the model's ability to position geographical locations and multitemporal RS tasks. To achieve this goal, the model needs to learn text embeddings that combine well with visual features so that it can be flexibly applied to multiple downstream tasks. Although some related MLLMs, such as RemoteCLIP [16] and RSGPT [3], have been proposed in the field of RS, several challenges remain as follows.

1) *Small-Scale Vision Language Pretraining Data:* The amount of open-source visual language data [8], [10], [17] is relatively limited, and usually only contains limited image and text data pairs, which is far from meeting the training needs of multimodal large models.
2) *Lack of Object-Level Information:* Existing RS image captioning datasets typically generate text descriptions derived from scene labels, which often lack detailed information about specific objects within the image, such as their location, quantity, and relationships. This limitation hampers their effectiveness in object-level tasks like object detection and grounded image captioning.
3) *Limited Application Fields of Existing MLLMs:* In practical applications, it is crucial not only to identify objects in static images but also to analyze scenes that change over time. However, most existing MLLMs primarily focus on static images and their corresponding text descriptions at a single point in time, lacking the capability to model and understand multitemporal information.

To address these challenges, we implemented a series of solutions. First, we created a large-scale, high-quality vision-language pretraining dataset consisting of 522 769 aligned image–text pairs. Unlike previous datasets that only describe the general scene of an image, our dataset includes detailed text descriptions that cover all objects of interest within the image, as well as additional specifics. This high-quality dataset is essential for effectively pretraining adapters, ensuring that they are more attuned to the unique feature representations specific to the RS domain. Second, we developed RingMo-VL1M, the largest instruction-tuning dataset in the RS field, comprising 1.6 million image–text pairs. Within RingMo-VL1M, we included a grounded image captioning dataset specifically designed for RS. By integrating data from various tasks—such as scene classification, VQA, object detection, image captioning, grounded image captioning, and change captioning—the model can learn from multiple instructions, thereby enhancing its performance across vision-language tasks. As illustrated in Fig. 1, we offer various tasks for users to engage with aerial imagery. Finally, leveraging both the pretraining and instruction-tuning datasets, we designed a foundation model that unifies grounding, vision, and language tasks in the RS domain. To improve the model's location accuracy and multitemporal analysis capabilities, we introduced a location- and instruction-aware querying transformer (Q-Former) and a change detection module. This comprehensive foundation model not only aligns image and text data but also learns object-level features and recognizes scene changes, making it highly effective for a wide range of RS applications.

In summary, our contributions can be summarized as follows.

1) *High-Quality Image–Text Pretraining Dataset:* We constructed a large-scale, high-quality image–text dataset through a novel generation process. Unlike existing image–text datasets, our pretraining dataset contains higher resolution images and more detailed text descriptions, making it particularly well-suited for the pretraining of RS foundation models.
2) *The Largest Instruction-Tuning Dataset With Multitemporal Images:* To ensure strong generalization across diverse tasks, we created an instruction-tuning dataset comprising 1.6 million question-answer pairs derived from various downstream tasks. These tasks include scene classification, object detection, VQA, image captioning, grounded image captioning, and multitemporal change captioning.
3) *A Unified Grounded Vision-Language Foundation Model:* We propose a unified foundation model that integrates vision, language, and grounded tasks. Through the design of specialized modules, this foundation model is equipped with capabilities for precise localization, instruction-aware processing, and multitemporal change analysis.
4) *Excellent Performance on 18 Datasets of Six Tasks:* The results across tasks like scene classification, object detection, image captioning, grounded image captioning, and change captioning demonstrate the versatility of the proposed foundation model. In addition, zero-shot evaluations on these tasks further confirmed the model's strong generalization capabilities.

II. Related Work

A. Instruction-Tuning Visual-Language Models

Instruction tuning is a technique designed to guide language models in following natural language instructions, which has been shown to enhance their generalization performance on novel tasks. Currently, there are two primary approaches to generating instruction data. The first approach utilizes rules or templates to create instruction formats from existing public datasets. The second approach leverages LLMs, such as the

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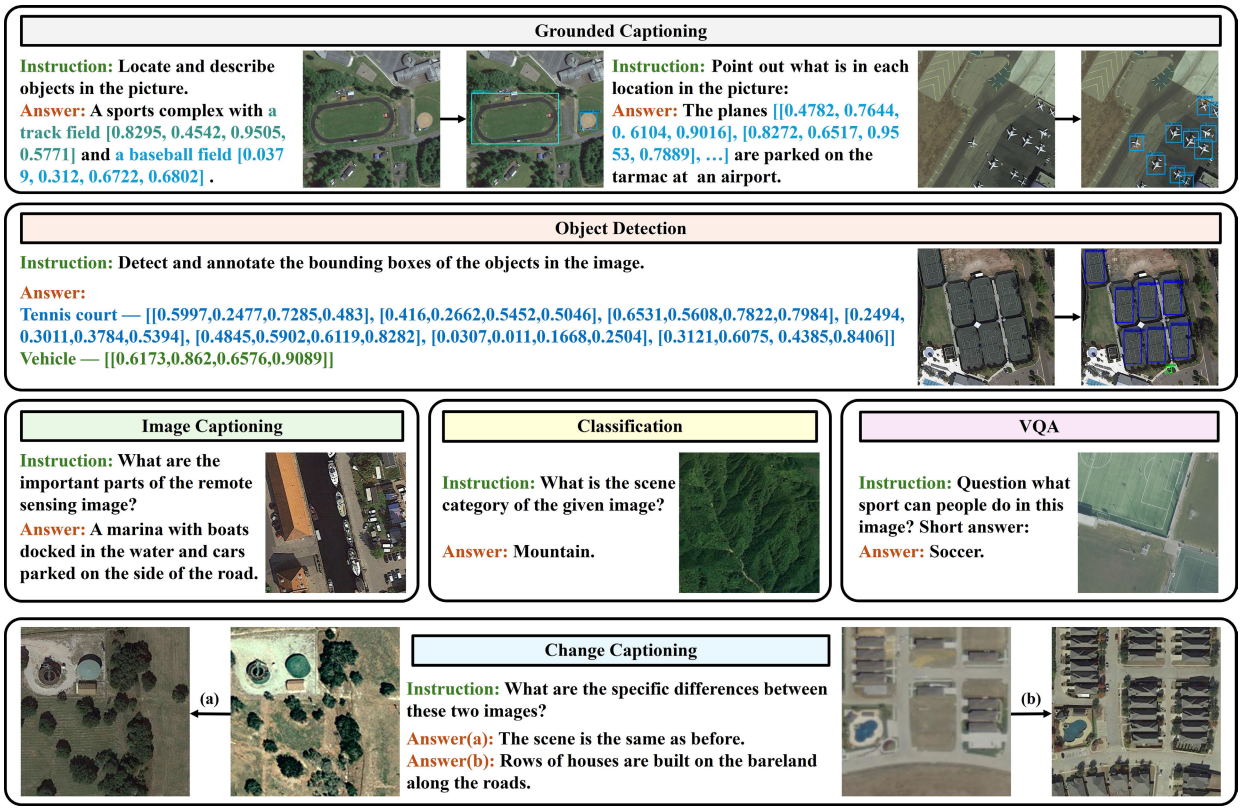


Fig. 1. RingMoGPT is a multimodal foundation model equipped with location- and instruction-aware capabilities. By following guided instructions, it can be applied to a wide range of tasks, including scene classification, object detection, image captioning, grounded image captioning, VQA, and change captioning.

GPT model, to generate more diverse and enriched instruction data.

Currently, many cross-modal large models combine visual models and LLMs to achieve the association and fusion of image and text information. Among them, instruction tuning has become a popular training strategy. For example, LLaVA [18] uses the language model GPT-4 [12] to extend the bounding box and captioning dataset in the COCO dataset [19] into a multimodal instruction-tuning dataset, which contains conversational QA, detailed captioning and complex reasoning data. InstructBLIP [14] converts 26 public datasets of 11 tasks into the format of instruction-tuning, and explores the zero-shot reasoning capability brought by instruction tuning. MiniGPTv2 [20] designs multitask instruction templates for instruction training. The Qwen-VL [21] model enhances command following and dialog capabilities through instruction-tuning, resulting in the interactive Qwen-VL-Chat model. Multimodal and text-only conversation data are mixed during the training process to ensure the model's versatility in conversational capabilities.

B. RS Visual-Language Models

Similar to natural scenes, there are many visual language tasks in the field of RS, such as image captioning, VQA, image retrieval, and visual grounding. Despite the progress made in these RS visual language models, they can only perform the specific tasks for which they were trained. The main reason is that these models do not have general semantic knowledge of RS images and cannot perform multiple rounds of conversation.

In RS, vision-language foundation models are designed to address a variety of tasks while maintaining conversational capabilities. RemoteCLIP [16] is the first vision-language foundation model in this domain, capable of handling tasks such as cross-modal retrieval, image classification, and object counting. Hu et al. [3] introduced the RSICap dataset to pretrain RSGPT, which contains 2585 human-annotated captions. While RSGPT demonstrates conversational abilities in image captioning and VQA, it requires fine-tuning for each task. GeoChat [22] pretrains an MLLM using instruction tuning. However, these models rely on pretraining with natural scene images for the instruction-tuning stage, neglecting the specific learning required for RS data during pretraining. RS-LLaVA [23] introduces an improved version of the large language and vision assistant model, specifically adapted for RS imagery through a low-rank adaptation approach. This model evaluates the performance of the model in the captioning and VQA tasks. SkyEyeGPT [24] is an RS vision-language model (VLM) capable of performing both single-task and multitask dialog. SkyEyeGPT freezes the visual encoder and employs the low-rank adaptation method to fine-tune the alignment layer and LLM. This model can handle a variety of tasks, such as image description, visual grounding, VQA, image captioning, referring expression generation, and scene classification. Although these approaches highlight the advantages of visual-language foundation models in RS, they primarily focus on scene-level tasks and offer limited interpretability for RS

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TABLE I
DETAILS ON THE SAMPLES USED FOR PRETRAINING, INSTRUCTION-TUNING, AND ZERO-SHOT STAGE

Stage	Category/Task	Dataset	Classes	Images/Samples	Image Size
Pretrain	Scene Classification	RSI-CB [26]	6	36309	256×256
	Scene Classification	RSSCN7 [27]	7	2800	400×400
	Scene Classification	AID [28]	30	9000	600×600
	Scene Classification	RSD46-WHU [29]	46	67589	256×256
	Scene Classification	PatternNet [30]	38	28800	256×256
	Scene Classification	OPTIMAL-31 [31]	31	1620	256×256
	Scene Classification	VArcGIS	38	55981	256×256
	Scene Classification	CLRS [32]	25	13200	256×256
	Scene Classification	NWPU-RESISC45 [33]	45	27300	256×256
	Scene Classification	Google/Bing	38	110512	-
	Object Detection	DOTA [34]	18	96798	800×800
	Object Detection	DIOR [35]	20	11923	800×800
	Object Detection	FAIR1M [6]	37	60937	800×800
Instruct	Scene Classification	AID [28]	30	10000	600×600
	Scene Classification	NWPU-RESISC45 [33]	45	31500	256×256
	Object Detection	DOTA [34]	18	11268	-
	Object Detection	DIOR [35]	20	23463	800×800
	Visual Question Answering	HRVQA [36]	27	1070240	1024×1024
	Change Captioning	LEVIR-CC [37]	-	50385	256×256
	Grounded Captioning	DOTA-GC	20	114477	400×400
	Grounded Captioning	DIOR-GC	18	35178	400×400
	Image Captioning	DOTA-Cap	20	250905	400×400
	Image Captioning	DIOR-Cap	18	35175	400×400
	Image Captioning	UCM-Captions [38]	21	10500	256×256
	Image Captioning	Sydney-Captions [38]	7	3065	500×500
	Image Captioning	RSICD [17]	30	54605	224×224
Zero-Shot	Scene Classfication	WHU-RS19 [39]	19	1005	600×600
	Scene Classfication	UCMerced-LandUse [40]	21	2100	256×256
	Object Detection	NWPU VHR-10 [41]	10	800	-
	Grounded Captioning	NWPU-GC	10	800	-
	Image Captioning	NWPU-Captions [10]	45	157500	256×256

objects. As a result, they are less suited for object-level tasks, particularly in multitemporal scenes.

III. DATASET

A. Large-Scale Vision-Language Pretraining Dataset

Using pretrained VLMs to generate captions for unlabeled RS images has become a popular method for addressing the scarcity of image–text paired data in RS. Inspired by RemoteCLIP [16] and RS5M [25], we combined the strengths of these two data generation approaches, leveraging popular public scene classification and object detection datasets to produce detailed textual descriptions and create a large-scale RS image–text pretraining dataset. Notably, compared to traditional RS image–text datasets, our dataset covers a broader range of scenes and object categories. The pretraining section in Table I provides detailed information about the data used.

After cropping and merging, we organized a total of 522 769 images, categorized into 67 scene categories and 25 object categories. By utilizing the category names from scene classification annotations along with the bounding boxes and category names from object detection datasets, we employed rule-based supervision prompts and the pretrained large VLM LLaVA-1.5 to generate image descriptions. Fig. 2 illustrates our data generation process and quality supervision methods.

1) Rule-Based Supervised Prompt: Similar to the RemoteCLIP method, we use a rule-based image labeling generation paradigm to create supervision prompts for each image. These prompts guide the VLMs in generating detailed image descriptions. For images from scene classification datasets, we use the scene annotation information, with the supervision text formatted as follows:

"The scene of ⟨category⟩."

For images from object detection datasets, we first resize the images to 800 × 800 pixels and then filter out images that do not contain objects. We then use the object detection labels to compile information on the object categories and their quantities within each image. The supervision text is generated using the format

"There is/are ⟨numbers⟩ ⟨category⟩ in image."

2) Multistyle Text Generation: Considering that a single sentence may not fully capture all aspects of RS images, we employ a multistyle prompting strategy to guide the VLMs in generating descriptions from four different styles.

Initially, we leverage the VLM's inherent sensitivity to the image by prompting it to provide both concise descriptions (e.g., "Describe this image simply in a confident tone.") and detailed descriptions (e.g., "Describe this image as detailed as possible.").

Subsequently, to address the limitation of existing description data that focuses solely on overall scene-level information, we use the standardized text generated in the first steps as prompts for the VLM. This approach directs the model to focus on specific aspects of the image, such as scene types and object information, thereby expanding its understanding of image details based on the supervision text.

Moreover, a step-by-step chain-of-thought approach is used to guide the visual language model in progressively deepening its understanding of the image, ultimately generating detailed descriptions. The complete template for the chain-of-thought prompts is shown in Fig. 2.

3) Annotation Evaluation and Correction Regeneration: To ensure the reliability of the generated annotations, we develop a self-assessment validation system. This system employs the pretrained RemoteCLIP [16] model to evaluate the image–text similarity of the generated text. We use the similarity scores obtained from the RemoteCLIP model on existing open-source image captioning datasets as a benchmark, comparing them with the similarity scores of the generated text. If the similarity score of the generated text is less than 80% of the benchmark score, or if there are significant discrepancies between descriptions of the same image, we will correct and regenerate these descriptions to enhance the accuracy and consistency of the annotations. The correction and regeneration process includes a small amount of manual sampling and relabeling, along with extensive self-checking and generation using large VLMs.

For manual sampling, we first group the organized images by category and then randomly select 10% of the data from each category for detailed inspection and correction. We systematically document and summarize the issues and defects identified during this process. Analysis indicates that the primary concerns are related to model hallucinations and subjective emotional statements. To address these problems, we design correction prompts to guide the large visual-language model in self-correcting and inspecting the remaining data. Through this process, we found that the large visual-language model performs exceptionally well in pseudo-label inspection and correction, significantly improving the quality of the generated annotations.

Ultimately, we generated four captions for each image from different perspectives. We obtained a total of 522 769 image–text pairs for the pretraining stage.

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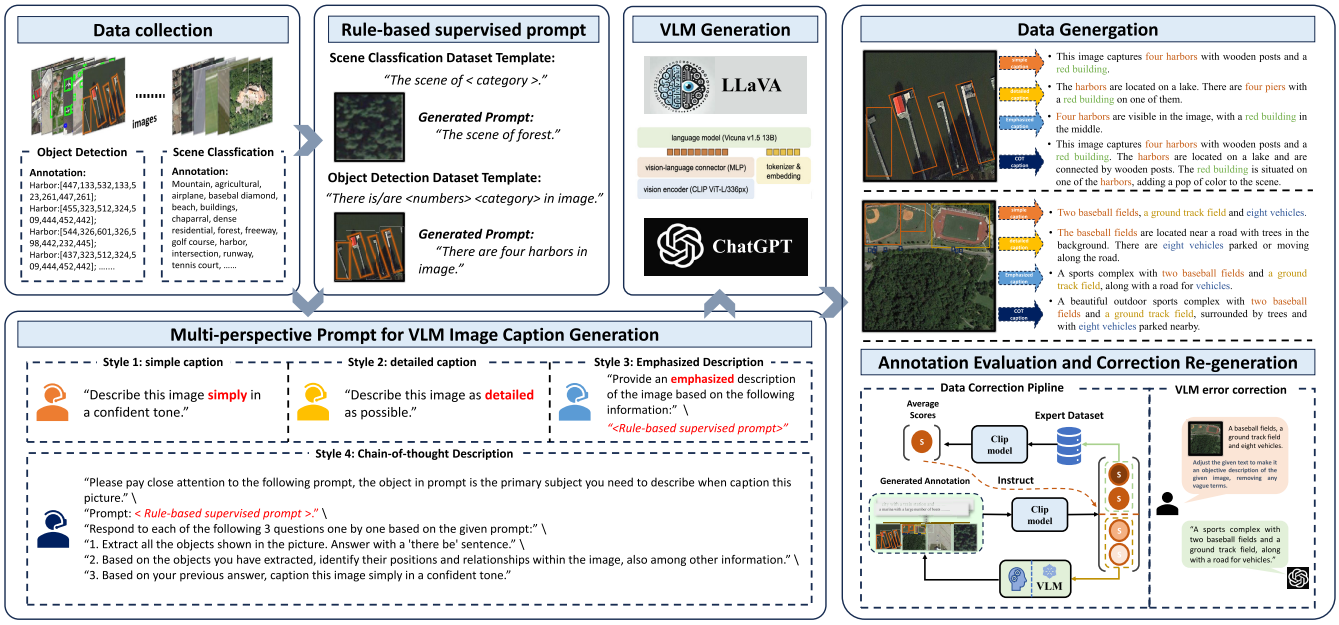


Fig. 2. Overview of the generation and validation process of our large-scale RS visual-language pretraining dataset: This process includes data organization, rule-based supervision prompt generation, diverse text generation, evaluation, and correction regeneration steps. We integrate public scene classification and object detection datasets, leveraging pretrained visual-language models to generate detailed image descriptions. Evaluation metrics consist of CLIP-based image–text similarity calculations and manual sampling. Identified issues are addressed through the MLLM's self-correction and regeneration process, along with manual reannotation.

B. Million-Level Vision Language Grounding Instruction Tuning Dataset

To enhance the model's ability to handle various tasks, we constructed an instruction dataset comprising over 1.6 million image–text question-answer pairs, as detailed in Table I. During the instruction-tuning phase, we comprehensively gathered both public and custom datasets, encompassing a broad spectrum of downstream tasks, including scene classification, object detection, image captioning, grounded captioning, VQA, and change detection.

Specifically, for image captioning and grounded captioning tasks, we created DOTA-Cap, DIOR-Cap, DOTA-GC, DIOR-GC, and NWPU-GC datasets based on the object detection datasets DOTA [34], DIOR [35], and NWPU VHR-10 [41]. The creation of DOTA-Cap and DIOR-Cap followed the method described in Section III-A, with images cropped to a size of 400 × 400 pixels. For the DOTA-GC, DIOR-GC, and NWPU-GC datasets, we initially employed the procedure outlined in Section III-A to generate image descriptions. Subsequently, we filtered the descriptions to retain only those that accurately included the object category labels. The remaining descriptions were then matched with object labels and bounding box information using a fuzzy matching mechanism. Finally, the integrated data underwent manual review to ensure the completeness and quality of the dataset.

For each task category, we design about ten instruction templates to enrich the diversity of questions. When generating the question-answer pairs, we randomly choose three out of the ten instructions as questions, which can increase the diversity of the instruction-tuning dataset.

IV. Methodology

A. Model Architecture

We introduce a grounded vision-language foundation model specifically designed for the field of RS. This framework incorporates an innovative Q-Former architecture, which is equipped with location- and instruction-aware capabilities. The model also includes a feature-matching module, a frozen LLM, a frozen visual encoder, and a change detection module. When trained using instruction-tuning, our model is capable of addressing both localization tasks and vision-language tasks simultaneously. The overall architecture of our method is depicted in Fig. 3. Below, we describe the main components in detail, with relevant symbols summarized in Table II.

1) Frozen Visual Encoder: We adopt existing vision models as visual encoders to reduce the cost of retraining. ViT-g/14 from EVA-CLIP [42] is employed as the visual encoder. The frozen visual encoder denotes as F_V takes an image (Img) as input, and extracts visual features as below

$$V = F_V(\text{Img}), \quad V \in \mathbb{R}^{N_V \times C} \tag{1}$$

where V denotes the extracted features from images. N_V and C are the output dimensions of the visual encoder.

2) Trainable Location- and Instruction-Aware Q-Former: Based on the image features processed by the frozen visual encoder V and the input instructions (Text), the Q-Former plays a crucial role in bridging visual and language tasks. It not only provides essential information to the frozen LLM but also locates and outputs significant instances within the images.

To achieve this, we establish two sets of learnable queries: semantic queries (Q_S) and location queries (Q_L). The semantic queries are designed to extract semantic features that align the visual encoder with the LLM. On the other hand, the

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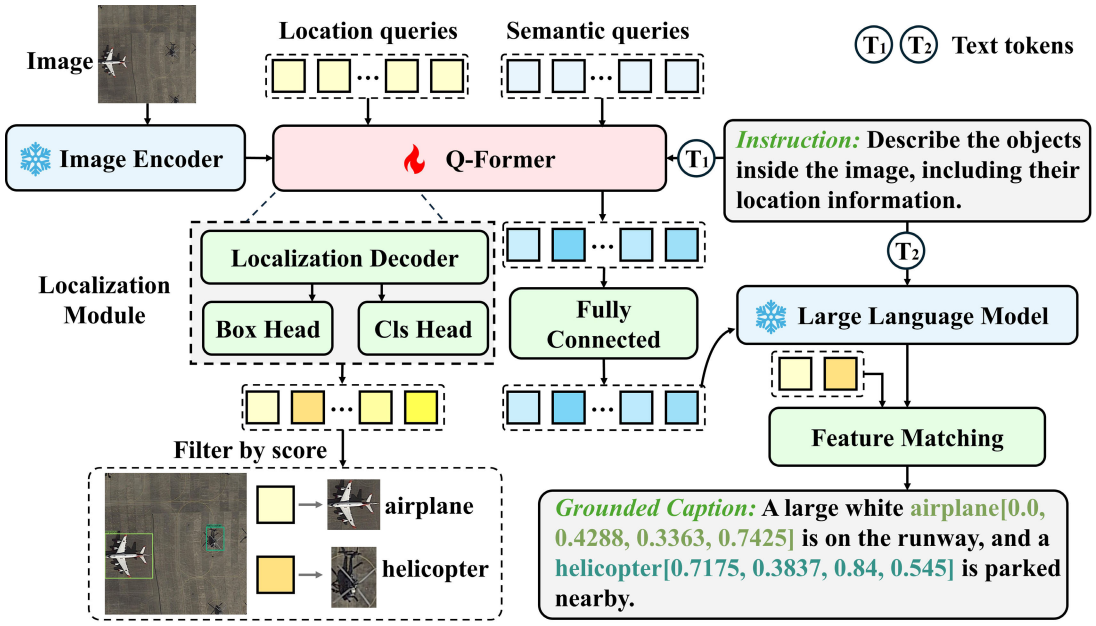


Fig. 3. RingMoGPT, an RS multimodal foundation model with cognitive capabilities. The model unifies various localization and vision-language understanding tasks under instructions from various tasks, by enabling location queries and semantic queries to learn and interact.

TABLE II
TABLE OF SYMBOLS, THEIR DIMENSIONS, AND MEANING

symbol	meaning	dimension
Img	Image	
$Text$	Text/Instruction	
F_V	Frozen visual encoder	
F_L	Frozen large language model	
V	Extracted features from the visual encoder	$N_V \times C$
V_1	Extracted features in the change detection module	$N_V \times C$
V_2	Extracted features in the change detection module	$N_V \times C$
Q_S	Semantic queries	$N_S \times D^V$
Q_L	Location queries	$N_L \times D^V$
f_V	Visual extraction module	
f_L	Localization decoder	
E_t^V	Text embeddings in the Q-Former	$N_{in}^V \times D^V$
E_t^L	Text embeddings in the large language model	$N_{in}^L \times D^L$
Q_L^V	Location queries with key visual features	$N_L \times D^V$
Q_S^V	Semantic queries with key visual features	$N_S \times D^V$
Q_L^L	Location queries with key location features	$N_L \times D^V$
R_{box}	Box head	
R_{cls}	Classification head	
S_{cls}	Output classification logits	$N_L \times 1$
S_{box}	Output coordinates of bounding boxes	$N_L \times 4$
G_{cls}	Truth classification	$N_G \times 1$
G_{box}	Truth coordinates of bounding boxes	$N_G \times 4$
S_{out}	Output sequence	N_{out}

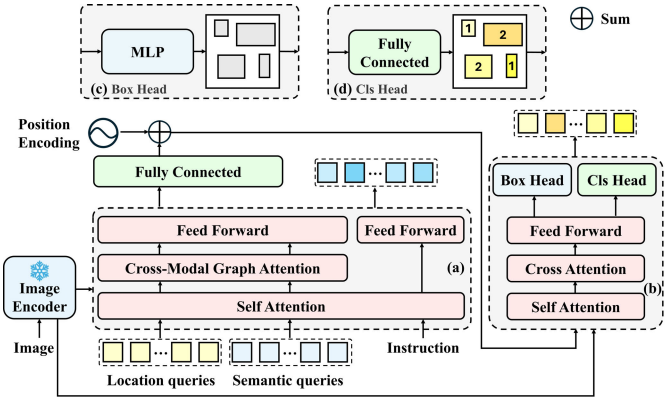


Fig. 4. Location- and instruction-aware Q-Former architecture. (a) Visual extraction module, responsible for extracting visual key features and aligning textual and visual features. (b) Localization module, which uses location queries to decode visual features, simultaneously predicts classes and bounding boxes. (c) Box head, tasked with predicting bounding boxes. (d) Classification head, responsible for predicting classes.

location queries focus on identifying and extracting objects of interest within the image, such as ships, airplanes, or other relevant entities that may be central to the task at hand.

The Q-Former architecture consists of two core components: a visual extraction module and a localization module, as illustrated in Fig. 4. The visual extraction module f_V is built from two transformer submodules that share the same self-attention layer. These include an image transformer, responsible for extracting key visual features, and a text transformer, which aligns both textual and visual features. To map words into a shared semantic space, we employ BERT [43] as our text encoder.

In RS, accurately extracting features from rigid structures like airplanes and buildings is critical. To address this challenge, we integrate graph attention networks (GATs) [44] into the cross-attention module. By doing so, we compute adjacency relationships between the hidden states of text and visual features, leveraging graph neural network models to propagate and update local information. This approach enhances feature representation and improves spatial context modeling.

To encode the text embeddings $E_t^V \in \mathbb{R}^{N_{\text{in}}^V \times D^V}$, each instruction corresponding to an image must first be encoded, where N_{in}^V is the number of tokens, and D^V denotes the hidden layer's dimension. The steps for visual feature extraction are outlined below

$$Q_L^V, Q_S^V = f_V(\text{Img}, E_t^V, Q_L, Q_S) \tag{2}$$

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where Q_S^V and Q_L^V represent the key features in the images queried after interacting with instructions. Q_S^V has learned the potential semantic representation of visual features, which will be further utilized in subsequent LLMs to accomplish language generation tasks.

The localization module consists of three main components: 1) a localization decoder f_L, which is made up of six transformer submodules to handle decoding tasks; 2) a box head R_{box}, responsible for predicting four coordinates that define the bounding box; and 3) a classification head R_{cls}, which outputs the semantic categories. To incorporate positional information into the input sequence, position encodings are introduced into Q_L^V. After passing through a linear projection layer, Q_L^V is mapped to the dimensional space required for the decoding layers. The box head R_{box} is implemented using a three-layer multilayer perceptron (MLP), while the classification head R_{cls} is designed as a simple linear projection layer.

Each output embedding from the localization decoder is then fed into R_{box} and R_{cls} to predict bounding boxes and classes, as follows:

$$Q_L^L = f_L\left(\text{Img}, Q_L^V\right) \tag{3}$$

$$S_{\text{cls}} = R_{\text{cls}}\left(Q_L^L\right) \tag{4}$$

$$S_{\text{box}} = R_{\text{box}}\left(Q_L^L\right) \tag{5}$$

where S_{cls} represents output classification logits and S_{box} represents output coordinates of bounding boxes. The DETR model [45] has demonstrated the viability and efficacy of this step. We further enhance its detection capability for RS instances in our framework.

3) Change Detection Module: As illustrated in Fig. 5, we have successfully devised the process for the change captioning task by integrating a change detection module at the back of the image encoder. This task can be trained alongside other visual language tasks within the overarching framework. For this specific task, since we utilize two images concurrently as inputs to the image encoder F_V, we obtain features V_1 and V_2, representing information before and after the alteration, respectively. After concatenating the two image features into a unified representation, they are fed into a dedicated MLP module. This module computes visual differences and eventually yields dimensions identical to those of other tasks, thereby facilitating joint training in subsequent stages.

4) Frozen LLM: We utilize the pretrained weights from Vicuna-13B [46], a high-parameter language model known for its strong text-fitting capabilities. The semantic query Q_S^V is integrated into the LLM F_L through a linear projection, which results in the corresponding encoded embeddings $E_t^L \in \mathbb{R}^{N_{\text{in}}^L \times D^L}$. Here, N_{in}^L refers to the number of text tokens, and D^L represents the hidden layer dimension, as defined below

$$S_{\text{out}} = F_L\left(E_t^L, Q_S^V\right), \quad S_{\text{out}} \in \mathbb{R}^{N_{\text{out}}}. \tag{6}$$

In this equation, S_{out} denotes the output sequences, and N_{out} indicates the number of output tokens.

5) Feature Matching Module: A matching mechanism is essential for aligning phrases with the instances predicted by our model, which operates using two separate branches for generating sentences and bounding boxes. To implement this, we first calculate embeddings for the predicted categories of each instance. We then assess the similarity between these embeddings and the words or phrases in the generated sentences, as well as the predicted categories. Given that the token embeddings in the LLM have been extensively tested and optimized, we select the word or phrase with the highest confidence score to establish the match.

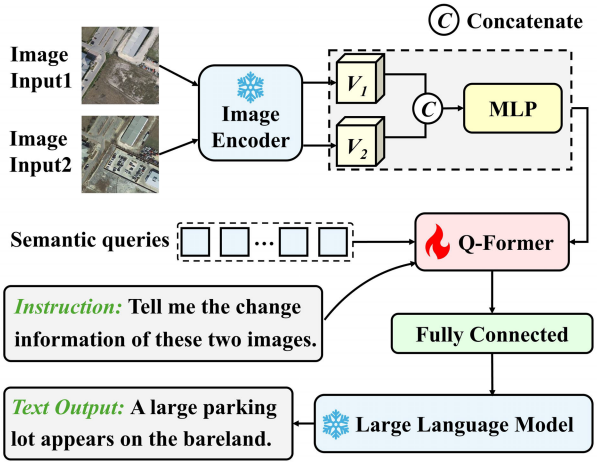


Fig. 5. Process of completing the change captioning task involves our model receiving input from two images simultaneously and extracting features in parallel. Crucially, it calculates significant change information within the change detection module, illustrated in the shaded area.

We initialize two sets of learnable queries: one set consists of $N_L = 100$ location queries, while the other comprises $N_S = 32$ semantic queries. Both sets of vectors have dimensions $D_V = 768$. The localization decoder utilizes a six-layer transformer architecture with a hidden layer size of 256. For decoding coordinates and categories, the model employs three MLP layers and a linear layer. In addition, the change detection module is constructed using two MLP layers to effectively handle changes in the input data.

B. Model Training

1) Pretraining Stage: We adhered to the training policy of BLIP2 [47] but introduced significant structural modifications to better suit the complexities of spatial features in RS images. Given these complexities, it became crucial to retrain the model using RS data, a process that greatly enhances the adaptability of our framework in both the change detection and localization modules. To achieve this, we pretrain the model with large-scale RS image–text pairs. The representation learning of images and text at this stage is depicted in the top section of Fig. 6. The loss functions during this phase employ various attention masks to regulate the output, following the methodology of BLIP2's first stage. Three loss functions are calculated: image-grounded text generation (ITG), image–text contrastive learning (ITC), and image–text matching (ITM). The objective of this stage is to develop deep representations of image–text pairs, thereby improving the model's comprehension of the relationship between text and images. The loss for this stage is formalized as follows:

$$L = L_{\text{itg}} + L_{\text{itc}} + L_{\text{itm}}. \tag{7}$$

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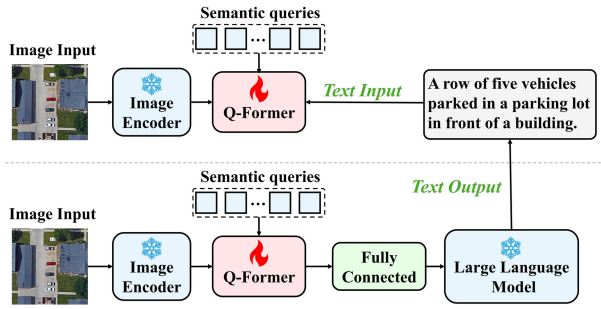


Fig. 6. Training process in the pretraining stage involves two main aspects. The representation learning of images and text (top), and the learning process of generating text from images (bottom).

Here, L_{itg} encourages the model to utilize context information to predict masked words. L_{itc} aims to maximize the mutual information between image and text representations by aligning them. To achieve this, it compares the image–text similarity of a positive pair with that of a negative pair. In addition, L_{itm} seeks to understand the precise alignment of text and image representations. The model is trained to predict whether a pair of images and texts is positive (matched) or negative (unmatched) in this binary classification task.

Next, we utilize RS image–text pairs to train the model's text generation capability, keeping both the image encoder and LLM frozen. By this stage, the semantic queries have already been pretrained and are proficient at extracting essential image features. We then align the dimensions of the semantic queries in the Q-Former with those of the LLM by mapping them through a fully connected layer. The bottom section of Fig. 6 illustrates the process of generating text from images. During this phase, the prediction loss L from the LLM is applied. The primary goal of this training phase is to further improve the model's visual language understanding and generation skills, particularly in generating text from images.

2) Instruction-Tuning Stage: The goal of this stage is to enable the previously unexplored location-aware function of Q-Former. The loss function at this stage comprises both language-task loss and localization loss. We employ the Hungarian algorithm [48] to determine the match between predictions and ground truth. The localization loss encompasses common losses from object detection tasks, including the classification loss L_{cls}, localization loss L_{loc}, generalized IoU loss L_{iou} [49], and auxiliary losses L_{aux}

$$L_{\text{det}} = \lambda_{\text{cls}} \times L_{\text{cls}} + \lambda_{\text{loc}} \times L_{\text{loc}} + \lambda_{\text{iou}} \times L_{\text{iou}} + \lambda_{\text{aux}} \times L_{\text{aux}} \tag{8}$$

where λ are hyper-parameters with values of 1, 5, 2, and 1, respectively. $L_{\text{cls}}(S_{\text{cls}}, G_{\text{cls}})$ denotes the cross-entropy loss computed using real and predicted labels, while $L_{\text{loc}}(S_{\text{box}}, G_{\text{box}})$ represents the L_1 loss computed with real and predicted coordinates. G_{box} and G_{cls} denote the coordinates and ground truth labels for each image, computed efficiently using predictions via the Hungarian algorithm. L_{aux} signifies the decoding loss in the localization decoder, whose effectiveness has been demonstrated in DETR [45].

The language task loss follows the prediction loss L_{llm} of the LLM, and the overall loss of the model can be written as:

$$L = L_{\text{llm}} + L_{\text{det}}. \tag{9}$$

C. Details

We conducted our training using eight A100 (80G) GPUs, employing the Adam optimizer with a weight decay of 0.05. The image resolution was set to 448×448 pixels. The learning rate was initialized at $1e^{-5}$ and then adjusted using a cosine decay schedule with a peak learning rate of $1e^{-4}$. In addition, a linear warm-up over the first 5000 steps was applied. The model was trained for 85 000 and 310 000 steps, respectively, in the initial phase.

To accelerate the fitting of location queries and enhance the model's localization capabilities, we implemented a specific strategy. Initially, only two localization-related tasks—object detection and grounded image captioning—were used in the fast-fitting training. At this stage, the initial learning rate was set to $4e^{-5}$, with a linear warm-up of 1000 steps and a peak learning rate of $1e^{-6}$. Subsequently, the model was trained for 300 000 steps with a configuration featuring an initial learning rate of $1e^{-5}$, a linear warm-up of 1000 steps, and a peak learning rate of 0.

V. Results

In this section, we evaluate our model's visual understanding capabilities across various vision-language tasks. After pretraining the model, we performed multitask fine-tuning on multiple datasets to assess its generalization abilities. To ensure a fair comparison with state-of-the-art methods, we also conducted specialized fine-tuning on each dataset and reported the best results. In addition, we introduce a novel baseline model that translates coordinates into text, built on the InstructBLIP framework [14]. In this study, we investigate the extension of the Q-Former into a language model capable of generating coordinate outputs while keeping the LLM frozen. For a more detailed design of the baseline model, please refer to the Appendix.

A. Image Captioning

1) Dataset: For the image captioning task, we utilize DOTA-Cap and DIOR-Cap along with three public image-caption datasets for instruction tuning and validation: UCM Captions [38], Sydney Captions [38], and RSICD [17]. In addition, we evaluate the model on the NWPU-Captions dataset [10] for a zero-shot test. In line with the settings of previous methods, we split the public datasets as follows: 80% of the image-caption pairs are used for training, 10% for validation, and the remaining 10% for testing.

2) Metric: Following the metrics in image captioning works, we apply METEOR, ROUGE-L, and CIDEr [59] as the metrics to evaluate the quality of generated sentences. METEOR measures the similarity between generated sentences and ground-truth sentences by considering factors such as word-level semantic similarity, synonyms, and word order.

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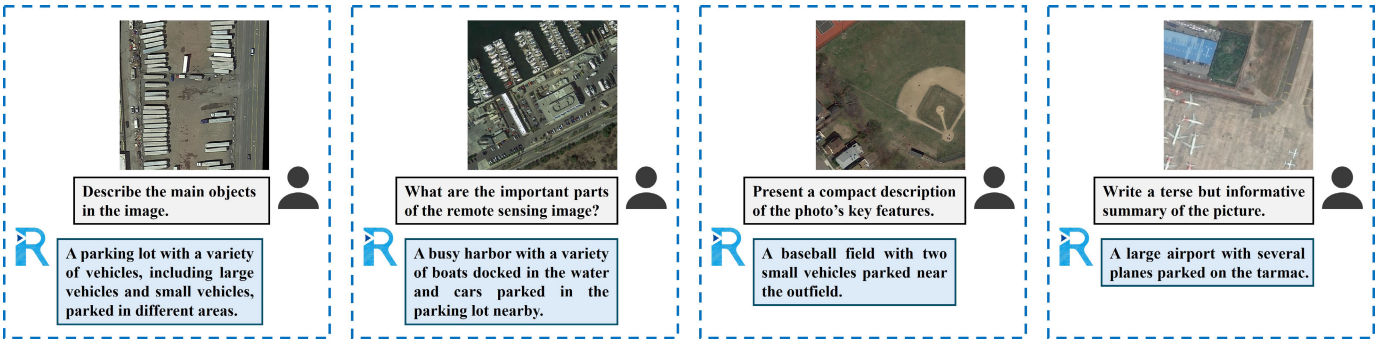


Fig. 7. Visualization results of the image captioning task. As can be seen from the figure, the sentences generated by RingMoGPT not only contain scene-level information, but also location association information between objects.

TABLE III
RESULTS ON THE DOTA-CAP, DIOR-CAP OF IMAGE CAPTIONING TASK

Methods	METEOR	ROUGE-L	CIDEr
Baseline	28.8	55.4	72.5
RingMoGPT	**30.4**	**60.1**	**99.2**

The ROUGE-L metric emphasizes the sequential information between generated sentences and ground-truth sentences. CIDEr adopts a consistency-based approach to evaluate the quality of generated sentences by comparing the similarity between multiple ground-truth sentences. These metrics evaluate the quality and accuracy of the sentences generated by the model.

3) Result: In this task, the model is provided with an RS image and tasked with generating a natural language description that thoroughly explains the scene, its features, or its contents. The generated text should include key information about the image. Some visualization results are shown in Fig. 7. We first evaluate RingMoGPT on datasets derived from DOTA-Cap and DIOR-Cap. By comparing the results with the baseline model, which relies solely on InstructBLIP, we observe that introducing location queries—capable of detecting important objects in the input image—leads to better performance. The key difference between RingMoGPT and the baseline method is that RingMoGPT leverages object location and spatial information for sentence generation. As shown in Table III, RingMoGPT clearly outperforms the baseline.

In addition, Tables IV–VI present the experimental results on three public datasets. As indicated in the tables, RingMoGPT consistently outperforms models specifically designed for image captioning. Our model achieves the best performance on the CIDEr metric and rarely includes descriptions of irrelevant objects. Despite being trained on multiple task datasets simultaneously, our results remain competitive, demonstrating that the use of instruction data enhances the model's image understanding capabilities.

4) Zero-Shot: To facilitate a more effective comparison of zero-shot performance among different vision-language foundation models, we selected several other well-established, high-performing models for experimental benchmarking. The results on the NWPU-Captions dataset are presented in Table VII. RingMoGPT significantly outperforms other large MLLMs, demonstrating remarkable zero-shot capabilities.

TABLE IV
RESULTS ON THE UCM-CAPTIONS TEST DATASET OF IMAGE CAPTIONING TASK

Methods	METEOR	ROUGE-L	CIDEr
CNN+RNN [50]	36.85	66.74	279.24
ConvCap [51]	28.31	59.62	190.15
Soft-attention [17]	38.86	72.37	261.24
CSMLF [52]	9.44	29.86	13.51
RTRMN [7]	42.58	77.26	312.70
Structure attention [9]	46.32	81.41	334.89
Word_Sentence [53]	43.95	71.32	278.71
SVM-D CONC [54]	37.02	68.77	292.28
GVFGA+LSGA [55]	44.36	78.45	332.70
MLCA-Net [10]	43.50	77.20	324.00
RASG [56]	45.71	80.72	338.87
CNN-T [57]	44.60	-	323.10
SCAMET [58]	**52.57**	81.66	337.72
RSGPT [3]	42.21	78.34	333.23
InstructBLIP [14]	45.50	80.23	330.94
SkyEyeGPT [24]	46.24	79.49	236.75
RS-LLaVA [23]	49.21	**85.78**	355.61
Baseline	48.34	82.82	351.18
RingMoGPT	49.90	83.29	**359.32**

TABLE V
RESULTS ON THE SYDNEY-CAPTIONS TEST DATASET OF IMAGE CAPTIONING TASK

Methods	METEOR	ROUGE-L	CIDEr
CNN+RNN [50]	30.36	57.47	168.94
ConvCap [51]	34.76	66.74	214.84
Soft-attention [17]	39.42	71.27	249.93
CSMLF [52]	24.75	50.18	75.55
Structure attention [9]	39.54	72.99	237.91
Word_Sentence [53]	41.81	69.22	204.11
SVM-D CONC [54]	36.43	67.46	222.22
GVFGA+LSGA [55]	38.66	70.30	245.22
MLCA-Net [10]	39.00	71.10	232.40
RASG [56]	39.08	72.18	263.11
CNN-T [57]	39.70	-	270.50
SCAMET [58]	46.14	72.18	235.70
RSGPT [3]	41.37	74.77	273.08
InstructBLIP [14]	38.70	70.80	250.52
SkyEyeGPT [24]	**46.62**	**77.74**	181.06
Baseline	40.85	74.41	267.79
RingMoGPT	42.07	73.39	**288.80**

We believe this improvement is due to RingMoGPT's specialized training on RS data, in contrast to generic MLLMs, which are trained on broader datasets. This domain-specific training provides the model with a deeper understanding, allowing it to capture granular, instance-level information. Consequently,

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TABLE VI
RESULTS ON THE RSICD TEST DATASET OF IMAGE CAPTIONING TASK

Methods	METEOR	ROUGE-L	CIDEr
CNN+RNN [50]	30.36	57.47	168.94
ConvCap [51]	33.25	57.70	166.48
Soft-attention [17]	32.55	61.09	196.43
CSMLF [52]	16.93	37.89	33.88
RTRMN [7]	28.29	55.39	151.46
Structure attention [9]	32.91	57.06	170.31
Word_Sentence [53]	31.97	62.60	206.29
SVM-D CONC [54]	22.99	45.57	68.54
GVFGA+LSGA [55]	32.85	59.29	260.12
MLCA-Net [10]	35.10	64.60	235.60
RASG [56]	36.26	66.91	275.49
CNN-T [57]	28.50	-	240.40
SCAMET [58]	**45.72**	**69.79**	246.81
RSGPT [3]	30.10	53.34	102.94
InstructBLIP [14]	27.42	50.25	93.80
SkyEyeGPT [24]	35.35	62.63	83.65
Baseline	33.52	59.91	254.20
RingMoGPT	34.26	61.62	**275.77**

TABLE VII
RESULTS ON THE NWPU-CAPTIONS ZERO-SHOT DATASET OF IMAGE CAPTIONING TASK

Methods	METEOR	ROUGE-L	CIDEr
Qwen-VL [21]	12.60	26.24	24.64
MiniGPTv2 [20]	13.70	26.60	19.70
LLaVA [18]	13.70	28.80	32.60
Baseline	18.81	34.43	48.08
RingMoGPT	**20.90**	**37.50**	**74.70**

RingMoGPT not only enhances recognition performance but also achieves significant gains in various RS tasks.

B. Scene Classification

1) Dataset: For the scene classification task, we use the AID dataset [28] and NWPU-RESISC45 dataset [33] as the training set and test set for instruction-tuning. This instruction-tuning dataset contains more than 110 000 images. In addition, we conduct a zero-shot evaluation using two complete public scene classification datasets: UC Merced Land-Use [40] and WHU-RS19 [39].

2) Metric: In this task, the model's objective is to accurately classify RS images into predefined scene categories. The model achieves this by predicting the correct category label for each image. A positive example is generated if the output text contains the true class, determined by searching and evaluating the model's output. We use accuracy as the evaluation metric, calculated as the ratio of correctly classified images to the total number of images.

3) Result: Table VIII shows the results of different methods under the training ratios (TR) of 20%. We provide queries concerning the scene classification of images, and based on whether the model correctly identifies the category, we compute the accuracy. To ensure fairness, apart from instructions, we did not provide any query candidate words to the model. Compared to other visual models, our model still yields competitive results. Fig. 8 shows output examples of scene classification in the test dataset.

4) Zero-Shot: Impressive zero-shot performance is achieved by our model, surpassing other visual language models.

TABLE VIII
RESULTS ON THE INSTRUCTION-TUNING TEST DATASET OF SCENE CLASSIFICATION TASK

Method	Year	AID (TR=20%)	NWPU-RESISC45 (TR=20%)
DCNN [60]	2018	90.82	91.89
CBAM [61]	2018	94.66	94.26
MSCP [62]	2018	92.21	90.81
RTN [63]	2018	92.44	92.71
CapsNet [64]	2019	91.63	89.18
SCCov [65]	2019	93.12	92.10
KFBNet [66]	2020	95.50	95.11
MG-CAP [67]	2020	93.34	92.95
EAM [68]	2021	94.26	94.29
F^2BRBM [69]	2021	96.05	94.87
MBLANet [70]	2021	95.60	94.66
GRMANet [71]	2021	95.43	94.72
CTNet [72]	2021	96.25	95.40
LSENet [73]	2021	94.41	93.34
MGML-FENet [74]	2021	96.45	95.39
ESD-MBENet-v1 [75]	2021	96.20	95.50
ESD-MBENet-v2 [75]	2021	96.39	95.36
SeCo-ResNet-50 [76]	2021	93.47	92.91
RSP-ViTAEv2-S [77]	2022	96.91	95.60
RingMo [1]	2022	96.90	95.67
EarthGPT [78]	2024	-	93.84
Baseline	2024	96.50	95.21
RingMoGPT	2024	**97.94**	**96.47**

TABLE IX
RESULTS ON THE ZERO-SHOT DATASET OF SCENE CLASSIFICATION TASK

Method	WHU-RS19	UC Merced Land-Use
Qwen-VL [21]	83.51	62.90
MiniGPTv2 [20]	62.08	4.76
LLaVA [18]	74.52	68.00
CLIP [79]	87.50	–
RemoteCLIP [16]	94.66	–
GeoChat [22]	-	84.43
Baseline	89.36	80.05
RingMoGPT	**97.71**	**86.48**

classification results on the WHU-RS19 dataset and UC Merced Land-Use dataset are reported in Table IX. Due to RingMoGPT being a visual language model specialized in RS scenes, it exhibits a more acute perception for scene classification in the field of RS, demonstrating strong generalization ability compared to other models.

C. Visual Question Answering

1) Dataset: For the VQA task, we use the HRVQA [36] as the instruction-tuning dataset. HRVQA dataset is the first high-resolution VQA dataset in aerial images, and it contains 53 512 RS images and 1 070 240 QA pairs. We select the training set and test set of the HRVQA dataset for instruction-tuning. HRVQA contains ten types of questions. In addition to the common questions in natural images, such as number and yes/no, HRVQA also contains location, shape, color, areas, transportation, scene, and sports.

2) Metric: In this task, the model must output the correct answer to a given question. The model should be able to understand the question and give an accurate natural language response. Following the metrics in the VQA task, we use accuracy as the evaluation metric, which consists of overall accuracy (OA), average accuracy (AA), and accuracy of each type of question.

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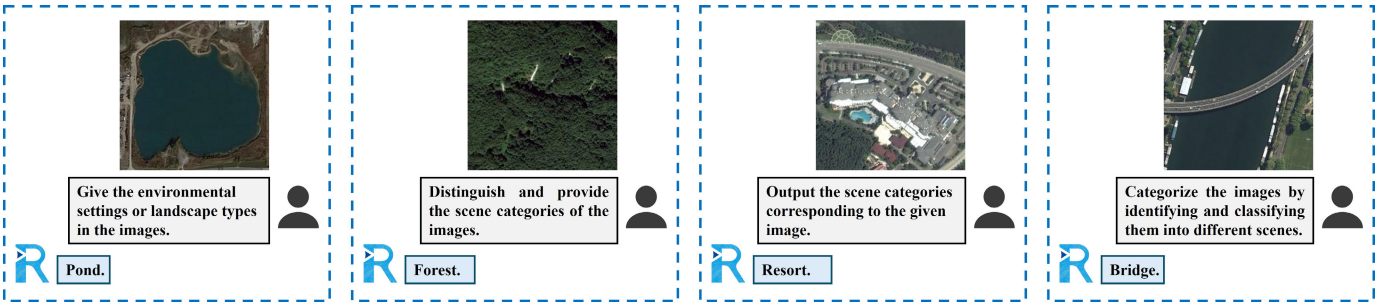


Fig. 8. Visualization results of scene classification tasks. The model can give the category of the scene based on the input image.

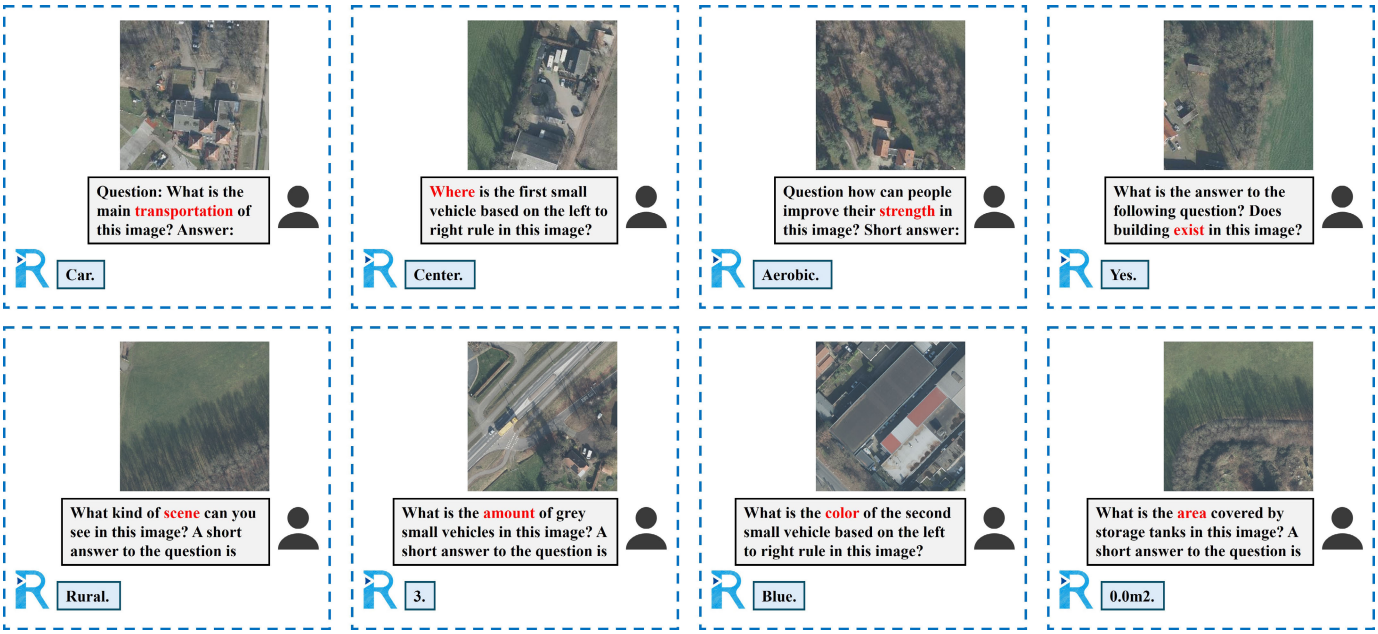


Fig. 9. Visualized results for VQA tasks. The model supports number and yes/no, location, shape, color, areas, transportation, scene, and sports types of questions.

TABLE X
RESULTS ON THE INSTRUCTION-TUNING TEST DATASET OF VQA TASK

Methods	Number	Y/N	Areas	Size	Locat.	Color	Shape	Sports	Trans.	Scene	OA	AA
SAN [80]	62.38	79.80	85.04	93.41	57.92	66.91	91.00	71.22	88.62	76.92	79.36	77.32
MUTAN [81]	63.87	86.88	86.05	93.53	59.46	69.72	93.31	65.15	80.11	73.70	77.68	77.18
MCAN [82]	65.75	92.58	98.21	90.88	58.38	41.56	96.37	77.82	89.52	75.71	79.96	78.68
RSVQA [83]	63.59	86.56	77.90	93.44	58.47	69.23	90.81	62.12	84.04	70.74	77.51	75.69
M-CA [84]	66.65	92.80	95.11	93.70	59.27	40.97	95.59	77.58	90.08	80.03	80.79	79.18
TRAR [85]	66.96	**93.72**	**98.57**	87.43	58.93	46.55	**96.63**	77.23	89.66	80.25	81.15	79.59
AOA [86]	66.93	93.25	97.41	93.48	60.50	45.14	96.55	78.36	89.75	79.73	81.42	80.11
FETH [87]	64.23	87.86	88.37	93.23	59.62	68.85	91.51	67.14	85.27	72.29	79.05	77.84
Swap-m [88]	63.45	89.88	93.42	91.27	56.81	43.55	93.55	74.32	87.27	76.93	78.56	77.04
GFT [36]	65.84	92.31	97.86	93.77	60.88	45.60	96.27	78.97	90.42	**81.79**	81.85	80.37
InstructBLIP [14]	63.96	88.42	63.91	86.85	57.43	69.00	77.08	75.00	85.22	58.00	71.49	75.43
Baseline	64.44	90.34	76.27	65.26	65.20	69.95	90.47	78.29	90.55	71.13	77.20	76.19
RingMoGPT	**75.30**	92.08	74.51	**98.84**	**72.68**	**73.99**	92.97	**85.62**	**95.74**	77.69	**85.39**	**83.94**

3) Result: Fig. 9 shows examples of image-question-answer triplets. Table X reports the results of RingMoGPT and state-of-the-art (SOTA) methods on the HRVQA dataset. The accuracy of these methods is obtained by fine-tuning models as a downstream task. Although special modules are designed in these methods, as shown in Table X, RingMoGPT achieves the best OA and AA on the HRVQA dataset. This result shows that the vision-language foundation model can surpass traditional fine-tuning models in VQA tasks. We can also clearly observe that RingMoGPT shows significant performances in answering questions related to numbers and sizes compared to traditional methods with remarkable improvements of 8.34% and 5.07%. This is mainly because RingMoGPT built on transfer learning from LLMs inherits excellent numerical understanding and mathematical reasoning capabilities from language models, thus performing remarkably well in answering questions related to numerical categories.

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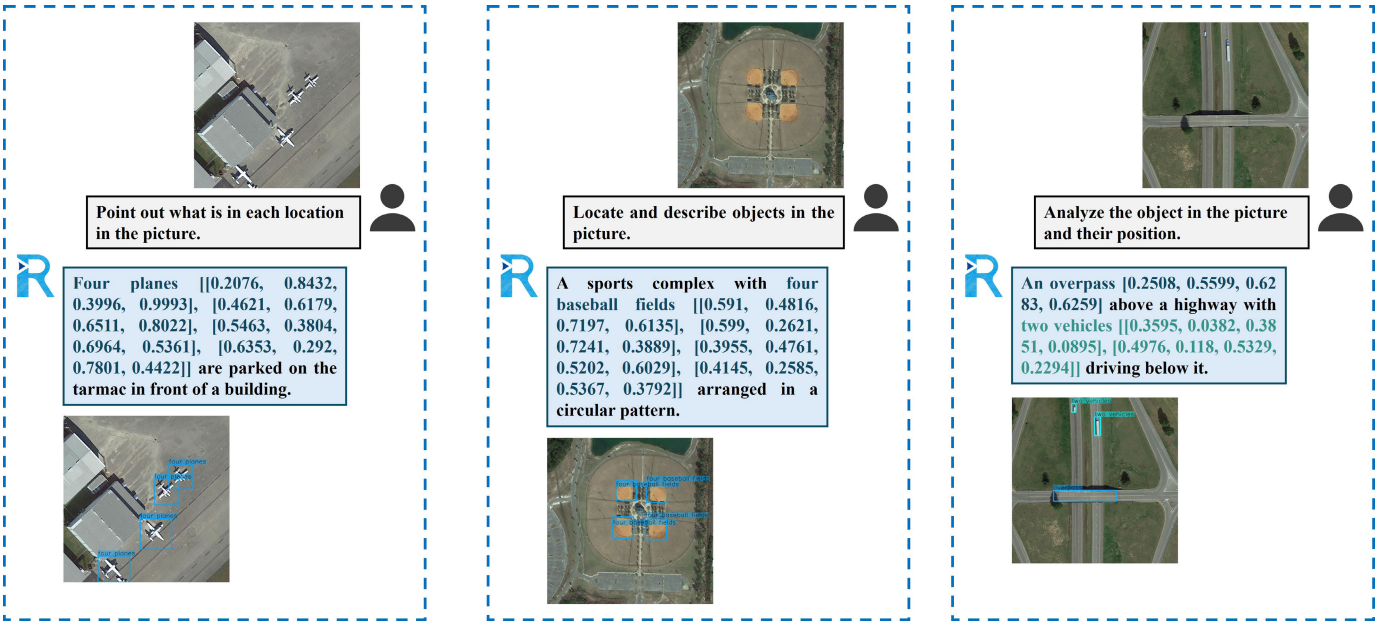


Fig. 10. Visualized results for grounded image captioning tasks. RingMoGPT can output the types of objects contained in the image and their location coordinates. RingMoGPT can generate high-quality descriptions of images and location coordinates of objects of interest.

TABLE XI
RESULTS ON THE INSTRUCTION-TUNING TEST DATASET OF GROUNDED CAPTIONING TASK

Methods	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	mAP(DIOR)	mAP(DOTA)
Baseline	67.5	54.2	43.6	34.8	28.7	**58.6**	92.7	44.9	35.6
RingMoGPT	**68.5**	**55.1**	**44.2**	**35.7**	**29.4**	57.3	**94.1**	**65.3**	**46.5**

TABLE XII
RESULTS ON THE ZERO-SHOT DATASET OF GROUNDED CAPTIONING TASK

Methods	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	mAP
MiniGPTv2 [20]	57.8	39.1	25.6	16.7	22.5	46.7	19.3	12.3
Qwen-VL [21]	48.1	30.6	19.4	12.6	11.8	32.8	26.5	6.34
Baseline	56.8	41.5	30.1	21.7	23.6	45.1	44.7	36.39
RingMoGPT	**58.8**	**45.2**	**34.9**	**27.0**	**25.3**	**48.4**	**47.7**	**52.73**

D. Grounded Captioning

1) Dataset: We constructed corresponding instruction-tuning data (DOTA-GC and DIOR-GC) and zero-shot evaluation data (NWPU-GC) based on the object detection dataset DOTA [34], DIOR [35], and NWPU VHR-10 [41]. In the grounded image–text pairs we constructed, the size of each image is 800 × 800, and each image corresponds to five ground-truth sentences, in which coordinate information is provided for all targets of the category of interest.

2) Metric: In this task, the model must produce natural language text that describes the contents of a designated area within the image. The text produced by the model should precisely and thoroughly depict the elements and settings of the designated area within the picture. As for the grounding performance, we use mean average precision (mAP) for evaluation, which is commonly used in object detection tasks. And we apply METEOR, ROUGE-L, BLEU-N ($N = 1$, 2, 3, and 4), and CIDEr as the metrics to evaluate the quality of generated sentences.

3) Result: Fig. 10 shows the visualized results for the grounded image captioning task. We assess our model on the instruction-tuning test dataset in Table XI, and this test dataset contains DOTA-GC and DIOR-GC. The experimental results show that we achieve promising results on the test dataset.

4) Zero-Shot: We also evaluate the grounded captioning performance of our method in the NWPU-GC test dataset, and the results are recorded in Table XII. Our method has achieved results comparable to other localization specialist models MiniGPTv2 [20] and Qwen-VL [21]. Especially, in terms of the object detection metric mAP, our model shows a significant improvement compared to MiniGPTv2 and Qwen-VL, reaching as high as 40%. This may be attributed to the extensive training RingMoGPT has undergone on RS images with rich and dense instances, which likely contributes to its impressive zero-shot performance in grounded captioning tasks.

E. Object Detection

1) Dataset: We use the DOTA [34] and DIOR [34] as the instruction-tuning datasets. NWPU VHR-10 [41] datasets are used in the experiments for zero-shot to test our generalization ability for detection tasks.

It is worth noting that to standardize the training and testing process for all object detection datasets, we adjust the vehicle

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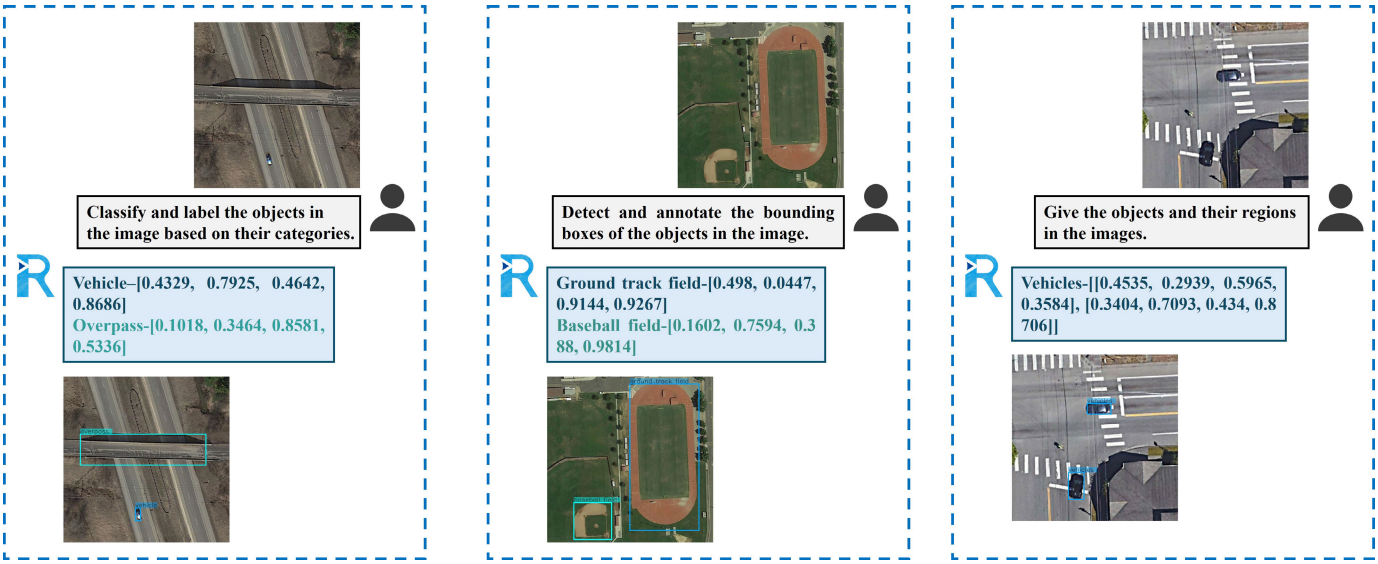


Fig. 11. Visualized results for object detection tasks. RingMoGPT can output the types of objects contained in the image and their location coordinates.

TABLE XIII
RESULTS ON THE DIOR TEST DATASET OF OBJECT DETECTION TASK

Methods	APL	APO	BF	BC	BR	CH	DAM	ESA	ETS	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
R-CNN [89]	35.6	43.0	53.8	62.3	15.6	53.7	33.7	50.2	33.5	50.1	49.3	39.5	30.9	9.1	60.8	18.0	54.0	36.1	9.1	16.4	37.7
RICNN [90]	39.1	61.0	60.1	66.3	25.3	63.3	41.1	51.7	36.6	55.9	58.9	43.5	39.0	9.1	61.1	19.1	63.5	46.1	11.4	31.5	44.2
RICAOD [91]	42.2	69.7	62.0	79.0	27.7	68.9	50.1	60.5	49.3	64.4	65.3	42.3	46.8	11.7	53.5	24.5	70.3	53.3	20.4	56.2	50.9
RIFD-CNN [92]	56.6	53.2	**79.9**	69.0	29.0	71.5	63.1	69.0	56.0	68.9	62.4	51.2	51.1	31.7	73.6	41.5	79.5	40.1	28.5	46.9	56.1
Faster R-CNN [93]	53.6	49.3	78.8	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	73.0	39.8	75.2	38.6	23.6	45.4	54.1
SSD [94]	59.5	72.7	72.4	75.7	29.7	65.8	56.6	63.5	53.1	65.3	68.6	49.4	48.1	59.2	61.0	46.6	76.3	55.1	27.4	65.7	58.6
YOLOv3 [95]	**72.2**	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	87.4	70.6	68.7	87.3	29.4	48.3	78.7	57.1
FPN-r50 [96]	54.1	71.4	63.3	81.0	42.6	72.5	57.5	68.7	62.1	73.1	76.5	42.8	56.0	71.8	57.0	53.5	81.2	53.0	43.1	80.9	63.1
FPN-r101 [96]	54.0	74.5	63.3	80.7	44.8	72.5	60.0	75.6	62.3	76.0	76.8	46.4	57.2	71.8	68.3	53.8	81.1	59.5	43.1	81.2	65.1
Mask-RCNN-r50 [97]	53.8	72.3	63.2	81.0	38.7	72.6	55.9	71.6	67.0	73.0	75.8	44.2	56.5	71.9	58.6	53.6	81.1	54.0	43.1	81.1	63.5
Mask-RCNN-r101 [97]	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	62.3	43.0	81.0	65.2
RetinaNet-r50 [98]	53.7	77.3	69.0	81.3	44.1	72.3	62.5	76.2	66.0	77.7	74.2	50.7	59.6	71.2	69.3	44.8	81.3	54.2	45.1	83.4	65.7
RetinaNet-r101 [98]	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
PANet-r50 [60]	61.9	70.4	71.0	80.4	38.9	72.5	56.6	68.4	60.0	69.0	74.6	41.6	55.8	71.7	72.9	62.3	81.2	54.6	48.2	86.7	63.8
PANet-r101 [60]	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1
CornerNet [99]	58.8	84.2	72.0	80.8	46.4	75.3	64.3	81.6	76.3	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9
CBD-E [100]	54.2	77.0	71.5	87.1	44.6	75.4	63.5	76.2	65.3	79.3	79.5	47.5	59.3	69.1	69.7	64.3	84.5	59.4	44.7	83.1	67.8
CSFF [101]	57.2	79.6	70.1	**87.4**	46.1	76.6	62.7	82.6	73.2	78.2	**81.6**	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
SCRDet++(RetinaNet) [102]	64.3	79.0	73.2	85.7	45.8	76.0	68.4	79.3	68.9	77.7	77.9	56.7	62.2	70.4	67.7	60.4	80.9	63.7	44.4	84.6	69.4
SCRDet++(FPN) [102]	66.4	83.4	74.3	87.3	**52.5**	78.0	70.1	84.2	78.0	80.7	81.3	56.8	63.7	73.3	71.9	**71.2**	83.4	62.3	**55.6**	**90.0**	73.2
FPN-RingMo(Swin-200W-200E) [1]	65.7	87.7	79.3	87.2	**52.5**	**80.5**	73.1	**91.0**	**80.8**	**84.5**	80.8	**64.8**	**66.1**	**88.7**	**75.9**	69.5	**87.7**	70.0	44.9	88.2	**75.9**
Baseline	56.7	63.3	62.8	25.4	34.0	64.3	59.8	44.0	41.7	51.7	48.8	37.8	43.9	21.3	66.9	21.1	40.1	69.2	13.4	73.2	47.0
RingMoGPT	64.2	**93.7**	77.2	71.0	40.5	77.0	**86.0**	69.3	53.9	77.1	61.8	54.5	55.0	49.5	38.5	45.1	67.2	**92.1**	34.7	85.1	64.7

* The abbreviations for category names are airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), expressway service area (ESA), expressway toll station (ETS), dam (DAM), golf field (GF), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM).

TABLE XIV
RESULTS ON THE DOTA TEST DATASET OF OBJECT DETECTION TASK

method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	Air	Heli	mAP
RetinaNet [98]	71.86	48.69	42.20	53.12	41.16	45.64	55.90	77.74	56.14	52.00	37.68	51.46	53.27	57.51	46.76	15.66	67.76	12.97	49.31
RetinaNet OBB [98]	70.99	46.77	43.76	55.08	41.55	51.06	58.01	77.78	57.72	53.50	37.66	51.72	53.56	57.32	49.18	12.01	64.53	4.48	49.26
MR [97]	77.61	51.35	44.89	60.12	42.51	48.10	57.93	77.84	57.55	57.88	36.53	51.71	54.79	**58.93**	**60.01**	14.42	60.32	8.43	51.16
CMR [103]	78.12	47.89	46.43	57.80	**42.97**	48.23	59.11	78.19	57.17	58.88	37.42	51.32	53.66	58.07	55.50	17.67	67.37	1.81	50.98
HTC* [103]	**78.28**	47.95	45.70	59.95	**42.97**	48.70	59.14	78.58	55.91	58.77	37.75	52.46	53.34	58.64	55.53	9.78	66.67	5.74	50.88
FR [104]	76.14	49.93	44.97	57.80	42.40	47.86	57.76	77.70	56.57	58.65	39.24	52.60	54.94	58.92	56.62	12.88	61.64	6.24	50.71
FR OBB [34]	71.68	45.80	45.56	58.70	42.18	51.28	59.28	79.01	58.74	58.75	37.26	51.93	52.36	58.08	54.12	8.48	53.01	2.40	49.37
FR OBB+Dp [105]	71.58	47.68	46.16	60.48	42.34	52.55	59.48	79.07	59.61	58.46	35.35	**53.73**	53.12	58.33	52.06	12.56	59.76	6.38	50.48
FR H-OBB [34]	77.14	50.54	45.60	57.53	42.27	48.09	57.60	78.40	59.78	57.80	36.64	52.13	52.51	58.42	48.91	14.99	60.00	8.47	50.38
FR OBB+RT [106]	71.84	48.20	**47.84**	**63.94**	**42.97**	**54.79**	**60.74**	**82.88**	**63.51**	**58.89**	**40.63**	52.83	55.70	58.87	57.94	**27.04**	64.27	7.68	**53.37**
Baseline	45.69	40.05	32.73	40.86	16.71	16.71	16.00	66.63	46.15	34.66	28.03	39.10	41.36	20.14	35.64	3.97	71.89	13.40	34.91
RingMoGPT	72.03	**58.02**	35.34	50.01	27.73	27.73	41.25	76.20	38.93	38.03	33.02	25.74	**63.23**	48.45	44.79	0.52	**95.75**	**25.20**	45.54

* The abbreviations for category names are plane (PL), baseball diamond (BD), ground track field (GTF), small vehicle (SV), large vehicle (LV), bridge (BR), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), helicopter (HC), swimming pool (SP), container crane (CC), airport (Air) and helipad(Heli).

category. While the DIOR and NWPU VHR-10 datasets only contain the "vehicle" category, the DOTA dataset includes "large vehicle" and "small vehicle." Therefore, to avoid ambiguity during prediction, we merge the two-vehicle subclasses in the DOTA dataset and keep only the "vehicle" category.

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TABLE XV
RESULTS ON THE ZERO-SHOT DATASET OF OBJECT DETECTION TASK

Methods	APL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mAP
Qwen-VL [21]	48.73	0.00	5.92	35.95	17.71	48.39	**97.1**	10.58	26.67	**27.15**	31.82
MiniGPTv2 [20]	28.93	0.00	12.64	38.03	21.51	41.00	83.58	27.65	60.40	25.85	33.96
Baseline	27.42	7.60	23.67	30.11	27.16	42.26	80.10	25.05	25.28	5.46	29.41
RingMoGPT	**88.32**	**10.22**	**43.12**	**78.55**	**56.46**	**61.69**	89.12	**30.76**	**76.42**	2.50	**53.72**

* The abbreviations for category names are airplane (APL), ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), and vehicle (VE).

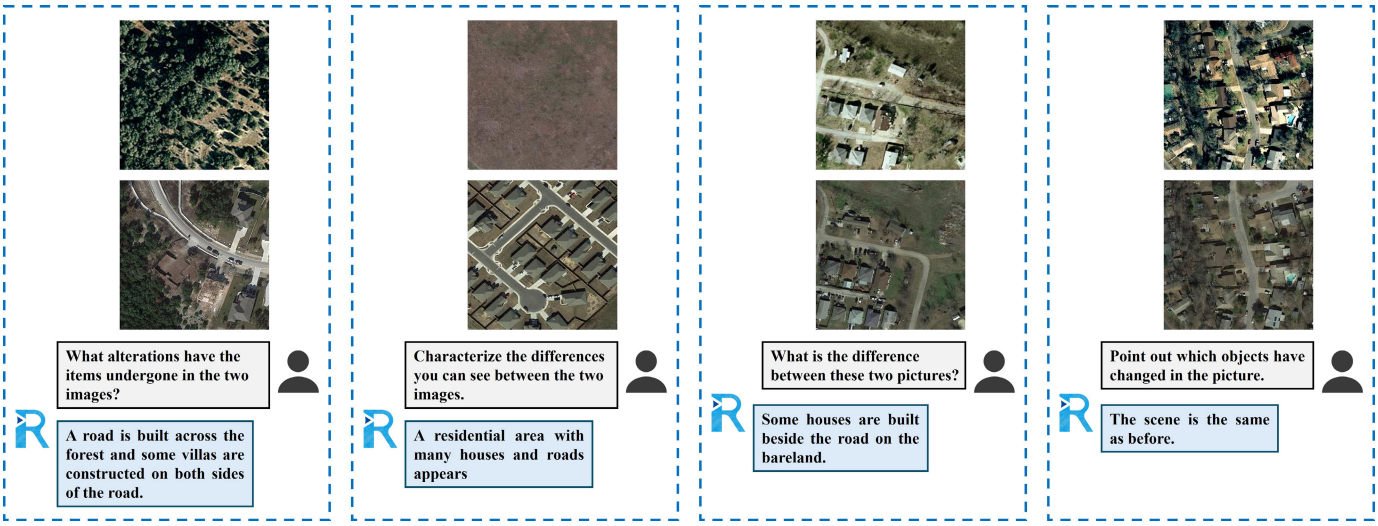


Fig. 12. Visualized results for change captioning tasks. RingMoGPT can generate descriptions of changing areas in two multitemporal images.

TABLE XVI
QUANTITATIVE EXPERIMENT RESULTS ON CHANGE CAPTIONING TASK

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
Capt-Rep-Diff [113]	72.90	61.98	53.62	47.41	34.47	65.64	110.57
Capt-Att [113]	77.64	67.40	59.24	53.15	36.58	69.73	121.22
Capt-Dual-Att [113]	79.51	70.57	63.23	57.46	36.56	70.69	124.42
DUDA [113]	81.44	72.22	64.24	57.79	37.15	71.04	124.32
MCCFormer-S [114]	79.90	70.26	62.68	56.68	36.17	69.46	120.39
MCCFormer-D [114]	80.42	70.87	62.86	56.38	37.29	70.32	124.44
RSICCFormer [37]	84.72	76.27	68.87	62.77	39.61	74.12	134.12
PSNet [115]	83.86	75.13	67.89	62.11	38.80	73.60	132.62
Pix4Cap [116]	**85.56**	**77.08**	**69.79**	**63.78**	39.96	**75.12**	**136.76**
Baseline	80.87	70.60	62.09	55.19	39.08	73.34	131.90
RingMoGPT	83.26	74.22	66.74	60.67	**40.25**	73.97	135.32

2) Metric: In this task, which focuses on RS images, the model must identify and locate different objects in the image. For each target that is identified, the model produces bounding box information and class labels that precisely identify the various objects present in the image. We compare each category's average precision (AP). The mAP is also examined.

3) Result: Tables XIII and XIV present the outcomes on two object detection datasets. On the DIOR dataset, our model demonstrates strong performance in identifying airports, dams, and train stations, achieving AP scores of 93.7%, 86.0%, and 92.1%, respectively. Similarly, on the DOTA dataset, our model excels in detecting bridges, harbors, airports, and helipads, with AP scores reaching 58.02%, 63.23%, 95.75%, and 25.2%, respectively. Several detection results are provided in Fig. 11. While our model may not achieve the highest performance levels compared to networks specifically tailored for object detection tasks (which may incorporate strategies like FPN [96] and Focalloss [98]), it attains comparable accuracy. This outcome underscores the potential of the vision-language foundation model in multicategory object detection tasks.

4) Zero-Shot: Our model is easily adaptable to various object detection without additional fine-tuning. we construct a zero-shot experiment on NWPU VHR-10 dataset, as shown in Table XV. The results show that our model can produce reliable results compared to other MLLMS.

F. Change Captioning

1) Dataset: For the task of change captioning, we utilize the LEVIR-CC dataset [37] for instruction-tuning. The LEVIR-CC

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TABLE XVII
INSTRUCTION TEMPLATE FOR EACH TASK

Task	Instrucion Template
Image Captioning	<Image>Describe the main objects in the image:
	<Image>What are the important parts of the remote sensing image:
	<Image>Present a compact description of the photo's key features:
	<Image>Write a terse but informative summary of the picture:
	<Image>Give a brief description of the scene in the image:
	<Image>Render a clear and concise summary of the photo:
	<Image>Share a concise interpretation of the image provided:
	<Image>Summarize the visual content of the image:
	<Image>Offer a succinct explanation of the picture presented:
	<Image>Please provide a short depiction of the picture:
VQA	<Image>{Question)
	<Image>Question:{Question)
	<Image>What is the answer to the following question?"{Question}
	<Image>{Question} The answer to this question is
Image Classification	<Image>Classify the given images into different scene categories:
	<Image>Output the scene categories corresponding to the given image:
	<Image>What is the scene category of the given image:
	<Image>Identify and label the scenes depicted in the images:
	<Image>Give the environmental settings or landscape types in the images:
	<Image>Categorize the images by identifying and classifying them into different scenes:
	<Image>Please output the scene corresponding to the image:
	<Image>Give the scene categories of the images:
	<Image>Distinguish and provide the scene categories of the images:
	<Image>Categorize the images based on their environmental setting or scene:
Object Detection	<Image>Detect and annotate the bounding boxes of the objects in the image:
	<Image>Classify and label the objects in the image based on their categories:
	<Image>Recognize and annotate the regions corresponding to different objects in the image:
	<Image>Detect and classify the objects in the image, providing both bounding boxes and class labels:
	<Image>Localize and annotate the regions of interest (ROIs) in the images:
	<Image>Identify and locate the instances in the images:
	<Image>Please output the object and their position in the images:
	<Image>Give the objects and their regions in the images:
	<Image>Detect entities in the images and mark their coordinates to indicate their locations:
	<Image>Locate objects and indicate their precise positions in the images:
	<Image>Find the objects and provide the coordinates of each object:
	<Image>Offer the objects and the position of each object in the images:
	<Image>Recognize objects and accurately locate them in the image:
Grounded Captioning	<Image>Which objects are in which part of the picture:
	<Image>Point out what is in each location in the picture:
	<Image>Represent the objects in the image and the corresponding locations of them:
	<Image>Locate and describe objects in the picture:
	<Image>Describe the objects inside the image, including their location information:
	<Image>Analyze the object in the picture and their position:
	<Image>Write all the objects in the picture and their position within one sentence:
	<Image>What can you notice in which region of the image:
	<Image>Where can you find the object in the picture and what are they:
Change Captioning	<Image>Please perform change caption in pictures:
	<Image>Analyze the differences in the given pictures:
	<Image>Point out which objects have changed in the picture:
	<Image>Characterize the differences you can see between the two images:
	<Image>Recognize changes before and after this area:
	<Image>What changes can be observed in the pictures:

dataset is derived from the LEVIR-CD dataset [107], which comprises 637 pairs of RS images, each sized at 1024 × 1024 pixels. To facilitate the description of changes, LEVIR-CC crops these original images into smaller patches of 256 × 256 pixels [108], [109] and annotates each patch with five distinct sentences. This results in a total of 10 077 pairs

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Single class and single object

Text Truth: **An empty airport runway surrounded by trees.**

Grounding Truth: **<grounding><phrase>airport</phrase><object><patch_0462><patch_index_3635></object>**

Single class and multi objects

Text Truth: **A factory with four large chimneys.**

Grounding Truth: **<grounding><phrase>chimneys</phrase><object><patch_0462><patch_1827></delimiter/><patch_0741><patch_2170></delimiter/><patch_1099><patch_1554></delimiter><patch_1747><patch_3373></object>**

Multi classes and multi objects

Text Truth: **Two harbors with two ships docked.**

Grounding Truth: **<grounding><phrase>ships</phrase><object><patch_1596><patch_1725></delimiter/><patch_2316><patch_2575></object><phrase>harbors</phrase><object><patch_1886><patch_2915></delimiter/><patch_2251><patch_2701></object>**

Fig. 13. Input representation of baseline method.

of effective image patches and 50 385 annotated sentences. In our experiments, we divide the dataset into 6815 samples for training, 1333 for validation, and 1929 for testing.

2) Metric: Following previous change captioning methods [37], [110], we adopt four quantitative metrics to evaluate the quality of generated sentences: BLEU-N ($N = 1, 2, 3,$ and 4), METEOR, ROUGE-L, and CIDEr. These metrics are widely used in image captioning tasks [111], [112] and measure the similarity between the reference and generated sentences. Higher scores on these metrics indicate better alignment and accuracy in the model's output.

3) Result: Table XVI presents the quantitative results for the change captioning task. While there is a performance gap compared to SOTA methods, RingMoGPT surpasses mainstream change captioning approaches like the Capt and MCCFormer series. Notably, compared to the baseline, RingMoGPT demonstrates significant improvements across all four evaluation metrics. We believe the reason RingMoGPT still trails behind some SOTA models is that those methods often employ carefully designed structures to capture differential information, whereas we rely on a simpler two-layer MLP to extract changes. Nonetheless, as shown in Fig. 12, RingMoGPT effectively mitigates the interference from pseudo-changes (such as shadows and lighting) and accurately describes the semantic changes between bi-temporal images.

G. Future Work and Limitations

While the proposed model demonstrates significant advancements in RS tasks, there are several areas that require further exploration and improvement as follows.

1) *Generalization Across Multisensor RS Data:* Our model performs strongly in tasks like image captioning, scene classification, and object detection, but it has primarily been trained on optical RS datasets. RS data encompasses a wide array of modalities, including radar, infrared, and multispectral imagery, which present unique challenges. Future work should focus on expanding the model's adaptability to these diverse data sources, improving its ability to generalize across different types of RS inputs.
2) *Computational Efficiency and Scalability:* Like many large-scale vision-language foundation models, our approach requires substantial computational resources for both training and inference. For practical real-world deployment, especially in resource-constrained environments such as edge devices or real-time applications, optimizing the model's architecture to reduce memory consumption and computational load will be crucial. Future efforts should focus on maintaining performance while improving efficiency and scalability.
3) *Fine-Grained Object-Level Interpretation:* Although the model performs well in scene-level tasks, interpreting fine-grained object-level information in RS images remains a challenge, particularly in complex environments where objects are small, overlapping, or occluded. Future research should aim to enhance object localization techniques, potentially refining the use of location queries or integrating more advanced segmentation methods, to better detect and interpret these fine-grained details.
4) *Limitations in Conversational Understanding:* While the model exhibits conversational abilities, there is room for improvement in understanding complex, multiturn dialogs, especially those requiring temporal or spatial reasoning. Future work should focus on refining the model's capacity to manage and respond to more intricate conversational scenarios, improving its utility in dynamic, real-world applications.

VI. Conclusion

We introduce RingMoGPT, a multimodal foundation model integrating vision, language, and localization functions. Leveraging domain adaptation, RingMoGPT efficiently trains

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by adjusting a small subset of parameters. To enable object detection and change captioning, we propose two novel components: Location- and Instruction-aware Q-Former and a dedicated change detection module. Meticulously designed pretraining and instruction-tuning datasets enhance RingMoGPT's performance. The pretraining dataset comprises over half a million high-quality image–text pairs generated through cost-effective methods. In addition, the instruction-tuning dataset includes over 1.6 million question-answer pairs across six downstream tasks. Experimental results demonstrate RingMoGPT's proficiency across all tasks, particularly in analyzing multitemporal data changes and identifying dense objects. Furthermore, RingMoGPT exhibits robust generalization capabilities under zero-shot data.

APPENDIX

A. Templates for Instruction-Tuning Data

Table XVII shows the templates of instruction-tuning data.

B. Setting of Baseline Method

In this process, we just removed the location queries and the location module, and used the learnable Q-Former to become the intermediate coordinate output structure. We maintain the answers of all tasks unchanged in the output, still optimizing the entire model through the generation loss of the LLM. Concurrently, we extended the original Bert model within the Q-Former framework to become a language model capable of midway prediction. In other words, we transformed the regression task of coordinates into the generation loss of a language model for computation.

For each class of the important instance in the image, there exist one or more bounding boxes representing them. We encode these bounding boxes into text and combine them with the instances. Specifically, we add a special token ⟨grounding⟩ at the beginning of the sentence to indicate the forthcoming coordinate prediction output. We use ⟨phrase⟩⟨/phrase⟩ as a set of special tokens to denote significant instances, while ⟨object⟩⟨/object⟩ is employed for one or more coordinates linked to the class. In addition, we divide the image into several patches, with each bounding box's four boundary points located in different patches. Each patch will be represented by a special token, and the generation process of coordinates follows the approach of the Kosmos-2 model [117]. When one category corresponds to multiple coordinates, the coordinates are connected using the special token ⟨/delimiter/⟩ as shown in Fig. 13.

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Peijin Wang (Graduate Student Member, IEEE) received the B.Sc. degree in automation from Tianjin University, Tianjin, China, in 2017, and the M.Sc. degree in signal and information processing from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2020, where she is currently pursuing the Ph.D. degree.

Her research interests include computer vision, deep learning, and remote sensing.

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Huiyang Hu (Graduate Student Member, IEEE) received the B.E. degree from Hebei University of Technology, Tianjin, China, in 2022. She is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

Her research interests include computer vision, remote sensing foundation models, remote sensing image interpretation, and self-supervised learning.



Boyuan Tong (Graduate Student Member, IEEE) received the B.E. degree from Central South University, Changsha, China, in 2022. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision and pattern recognition, especially on remote sensing image interpretation and multimodal signal processing.



Ziqi Zhang received the B.S. degree from the School of Electrical and Information Engineering, Tianjin University, Tianjin, China, in 2017, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing, China, in 2022.

He is currently a Research Associate with CASIA. His main research directions are multimedia learning, video captioning, and video retrieval. He has published four scientific papers in this area in top journals and conferences including IEEE TRANSACTIONS ON IMAGE PROCESSING, Computer Vision and Pattern Recognition (CVPR), and International Conference on Computer Vision (ICCV).



Fanglong Yao (Member, IEEE) received the B.Sc. degree from Inner Mongolia University, Hohhot, China, in 2017, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2022.

He is currently a Post-Doctoral Researcher and an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include cognitive intelligence, embodied intelligence, and swarm intelligence, concentrating on multiagent learning, multimodal fusion, 3-D scene understanding, and spatiotemporal data analysis.



Yingchao Feng (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2017, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2022.

He is currently an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and deep learning, especially on object detection, semantic segmentation, and remote sensing.



Zining Zhu (Graduate Student Member, IEEE) received the B.Sc. degree from Jilin University, Changchun, China, in 2020. She is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

Her research interests include computer vision, remote sensing image interpretation, few-shot learning, and incremental learning.



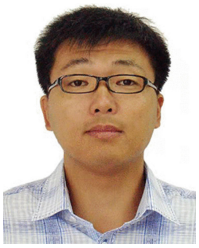
Hao Chang (Graduate Student Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include generative pretraining and remote sensing image understanding, especially on change detection and change caption.



Wenhui Diao (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, and concurrently at the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing. His research interests include computer vision and remote sensing image intelligent interpretation.



Qixiang Ye (Senior Member, IEEE) received the B.S. and M.S. degrees in mechanical and electrical engineering from Harbin Institute of Technology, Harbin, China, in 1999 and 2001, respectively, and the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2006.

He has been a Professor with the University of Chinese Academy of Sciences, Beijing, since 2009, and was a Visiting Assistant Professor with the Institute of Advanced Computer Studies (UMIACS), University of Maryland, College Park, MD, USA, until 2013. He has authored more than 100 articles in refereed conferences and journals including IEEE Computer Vision and Pattern Recognition (CVPR), International Conference on Computer Vision (ICCV), European Conference on Computer Vision (ECCV), IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON IMAGE PROCESSING, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY. His research interests include image processing, object detection, and machine learning.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.

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